SubCellSpace: A Subcellular Spatial Transcriptomics Database & Analysis Platform

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1 Platform Construction & Analysis Pipeline

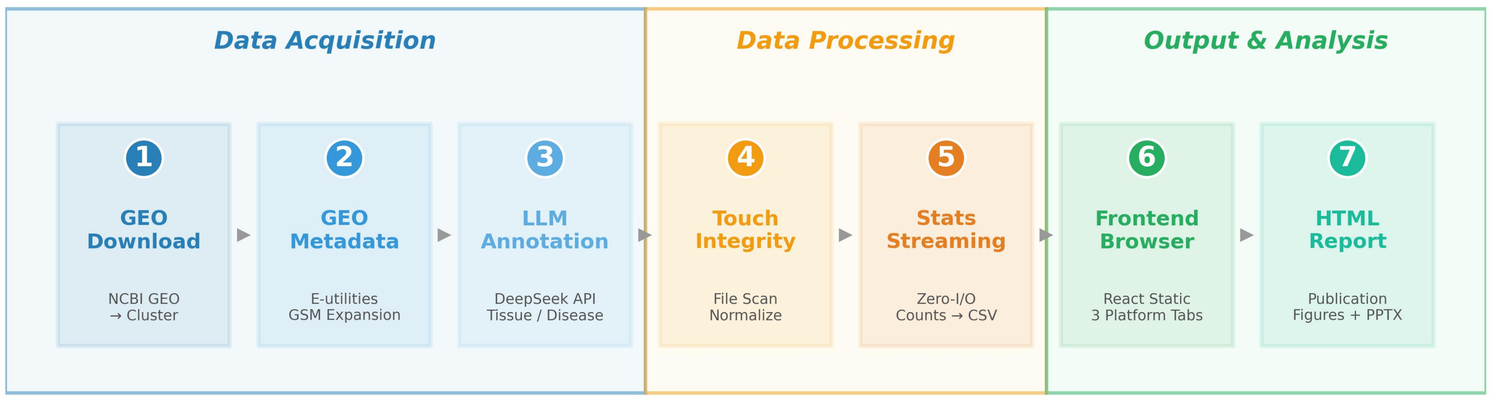
SubCellSpace is a unified subcellular spatial transcriptomics platform that supports CosMx (NanoString), Xenium (10x Genomics), and MERFISH (Vizgen) with a single end-to-end pipeline. The platform integrates three major phases:

1. Data Acquisition — GEO metadata curation via NCBI E-utilities (269 GSE series, 4,478 GSM records), LLM-powered tissue/disease annotation (DeepSeek), and automated file integrity checking through the Touch module.

2. Data Processing — Zero-I/O streaming statistics (Stats module) for transcript, cell, and gene counting. The 7-step analysis pipeline (illustrated) is executed via a single configuration file shared across all platforms.

3. Output & Visualization — React + TypeScript static frontend with platform-tabbed browsing, HTML report generation with publication-quality figures, and Slurm HPC deployment for batch processing (10 concurrent tasks).

Key innovations include the Touch module for automated file integrity normalization across naming inconsistencies (e.g., fov_position vs. fov_positions), and the Stats module for memory-safe streaming statistics with a 50,000-cell threshold. The database encompasses 2,410+ GEO entries across 269 GSE series, supplemented by 998 non-GEO public datasets — all uniformly processed through the same pipeline.



Analysis Pipeline — scverse-native 7-step workflow

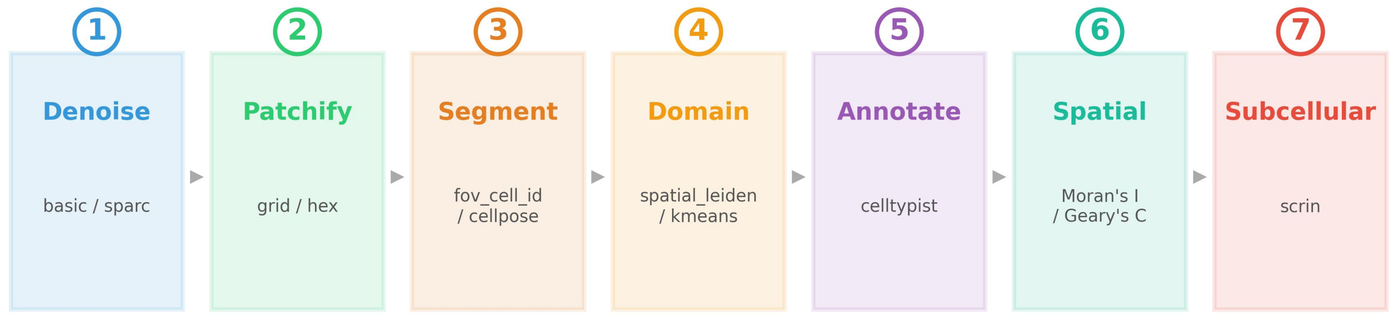


Fig 1. End-to-end platform construction pipeline (top) and scverse-native 6-step analysis workflow (bottom)

2 Database Overview: Composition, Coverage, Biology & Quality

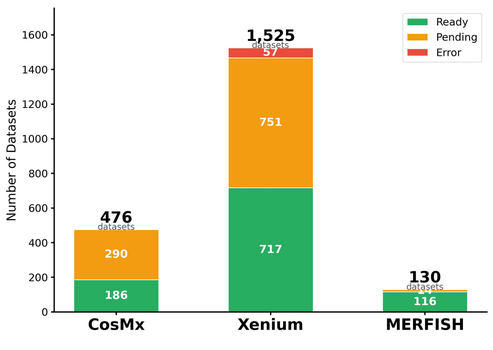


Fig 2. Dataset counts

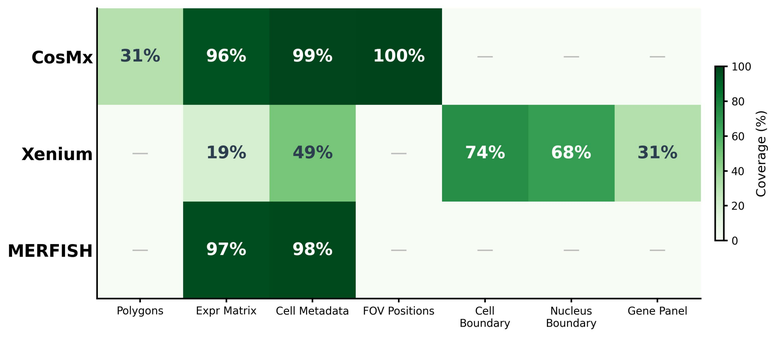


Fig 3. File coverage

Our database encompasses 2,410+ GEO entries across three major ST platforms. Xenium leads with 1,525 entries (717 ready). CosMx contributes 476 entries (186 ready) from targeted panels. MERFISH has 130 entries (116 ready), with the highest ready ratio.

File completeness varies: CosMx provides expression matrices (96%), cell metadata (99%), and FOV positions (100%). Xenium uniquely offers cell boundaries (74%), nucleus boundaries (60%), and gene panel metadata (31%). MERFISH excels in expression (97%) and cell metadata (98%). The Touch module auto-normalizes all.

The database spans 50+ tissue types and 30+ disease conditions. Brain, breast, lung, and lymph node are most represented, along with normal tissue, various cancers, and neurodegenerative conditions (Alzheimer's). Xenium achieves the highest transcript counts (181K median) and broadest gene panels (6,522). MERFISH captures the most cells (18K median, 550K max). CosMx offers consistent 960-gene panels with 35,825 per-cell transcript median.

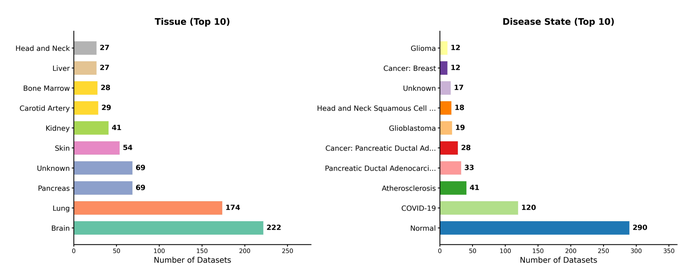


Fig 4. Tissues & diseases

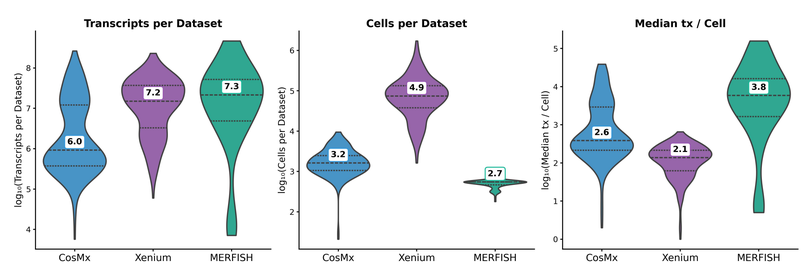


Fig 5. Quality metrics

3 Case Study: Alzheimer's Mouse Brain

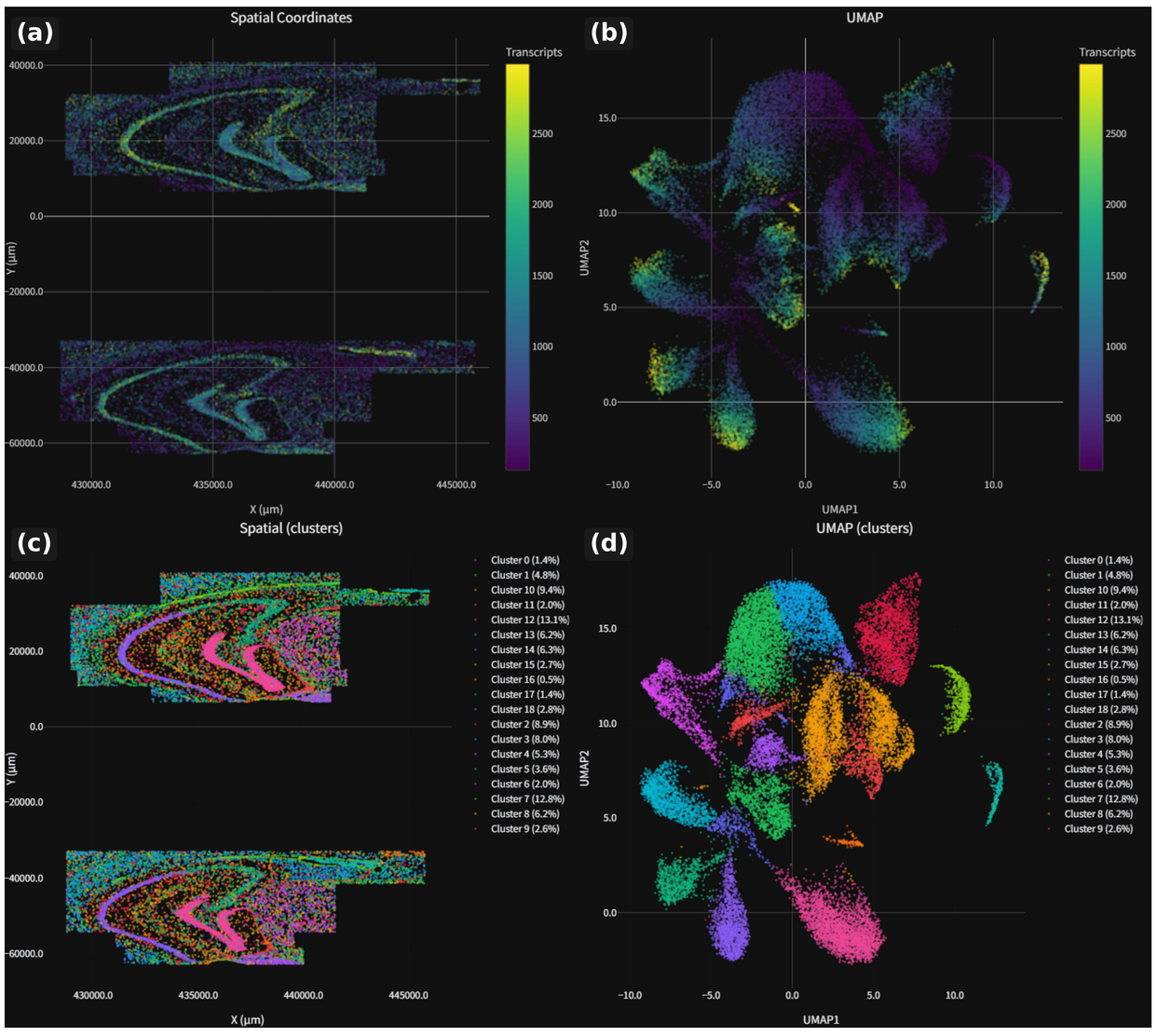
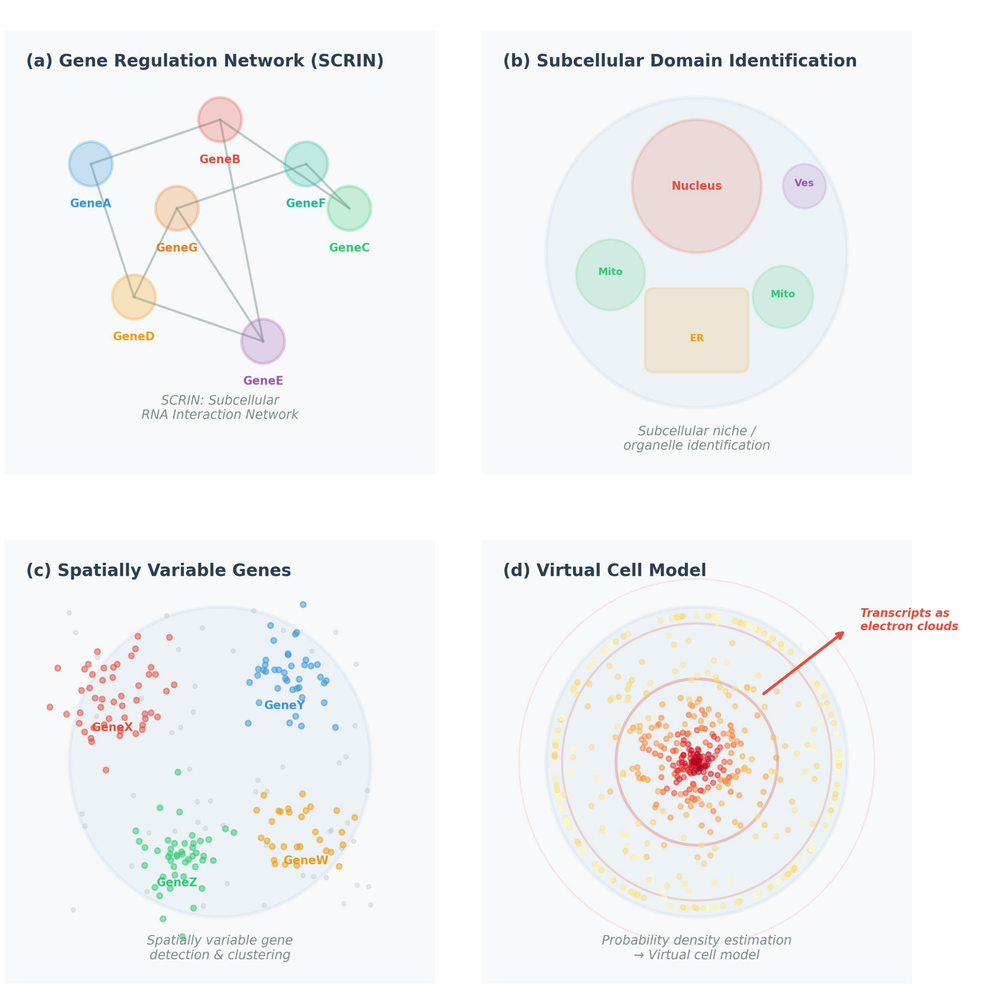


Fig 6. GSE263791 — Alzheimer's disease mouse brain (CosMx)

Alzheimer's disease mouse model (GSE263791). 42.6M transcripts, 821 cells, 960 genes, 41 FOVs. Median 35,825 transcripts per cell. (a) Spatial transcript distribution: hippocampal regional differences. (b) UMAP by transcript count. (c) Leiden clustering: 8 cell populations. (d) UMAP cluster validation. Microglia-astrocyte cross-talk in amyloid plaque niches revealed by spatial context.

4 Conclusion & Outlook



SubCellSpace delivers a unified database and analysis platform covering 2,410+ GEO entries across CosMx, Xenium, and MERFISH platforms. The database spans 50+ tissue types and 30+ disease states. The single-configuration pipeline enables cross-platform subcellular analysis through denoising, segmentation, domain detection, clustering, and annotation modules, complemented by SCRIN for gene regulatory network inference and spatial autocorrelation statistics. Future work will integrate spatial multi-omics data, deploy cloud-based analysis pipelines, and establish a community-contributed dataset registry with persistent identifiers to accelerate subcellular spatial transcriptomics research.